Ethernet Scale-Up Test Standard

LLR/CBFC Test Strategy and Latest Progress

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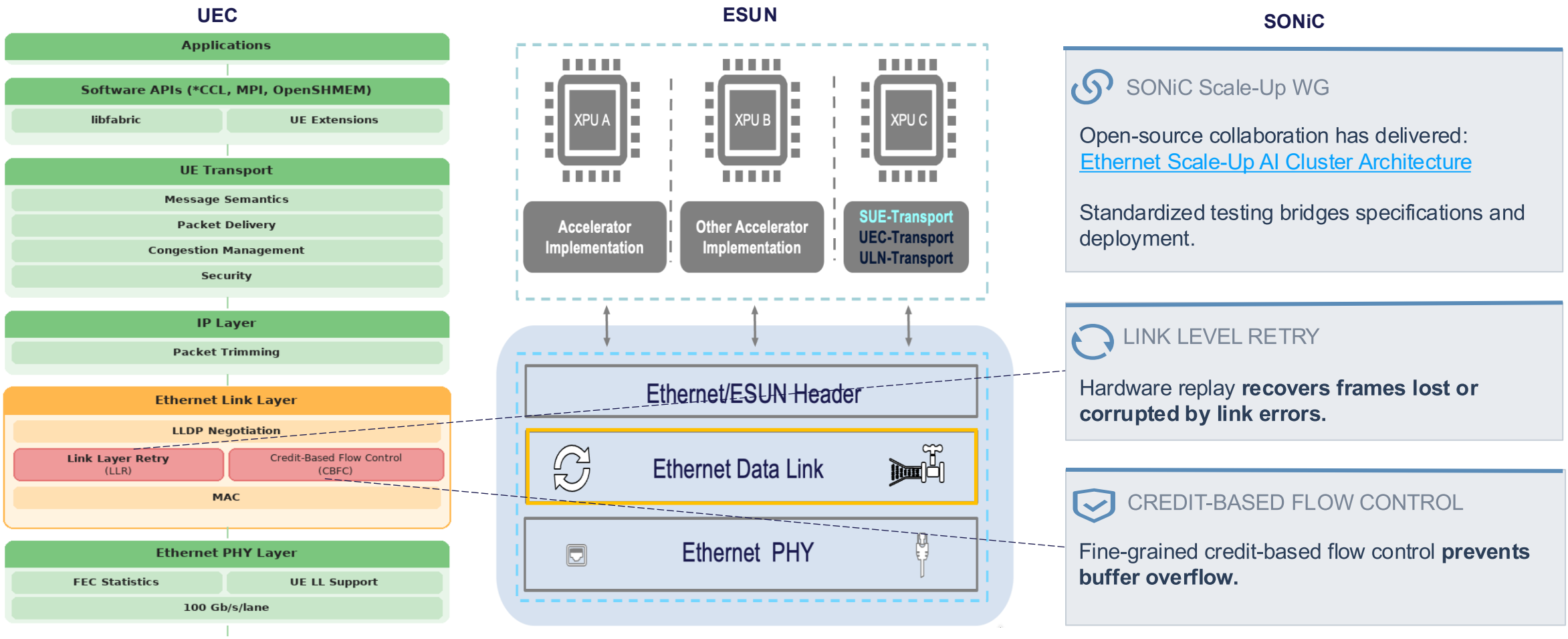
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LLR & CBFC - Overview and Deployment Challenges

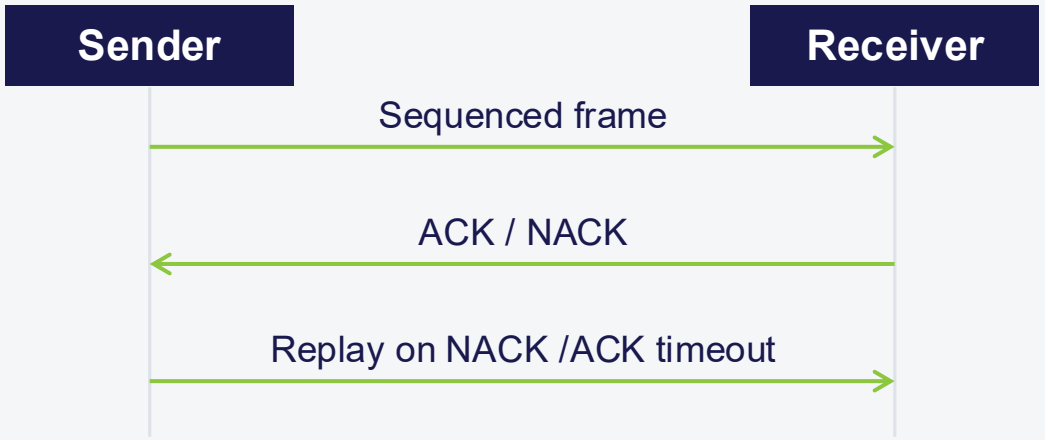
Deterministic, lossless networking for AI Scale-Up: from protocol specification to field deployment

Building deterministic and lossless networking for AI Scale-Up



LLR & CBFC Spec Overview

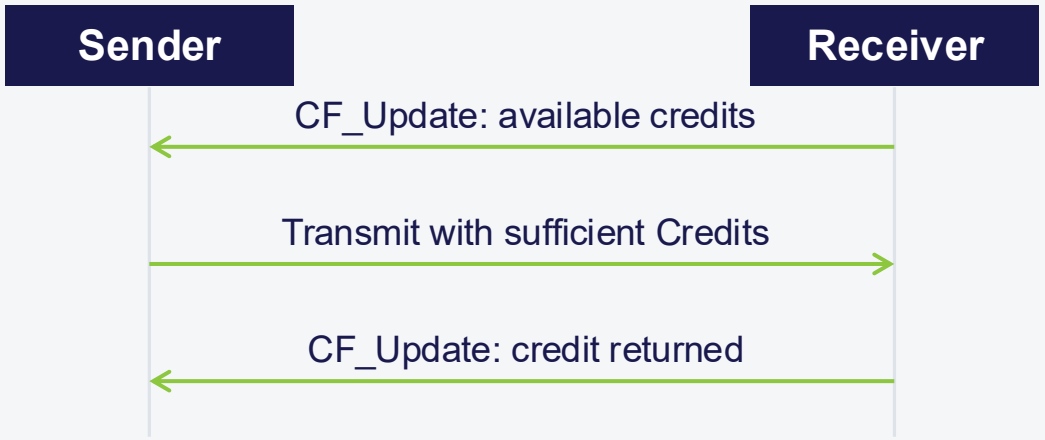
LLR — Link Level Retry



Sender Sequence and buffer frames until ACK; replay on NACK / ACK timeout.

Receiver Check integrity and sequence; return ACK/NACK; discard duplicate frames.

CBFC — Credit-Based Flow Control



Sender Track and debit credits per VC; pause when credits are insufficient.

Receiver Advertise buffer credits via CF_Update; return credits as buffer space is freed.

From Specification to Deployment: The Last Mile

Deployment Complexity

- Topologies and vendor differences
- RTT-dependent credit and replay-buffer sizing

Coupled Parameters

- Interdependent LLR / CBFC settings
- Misconfiguration can cause buffer overflow or limited throughput

Edge-Case Validation

- Poison CRC: store-and-forward vs. cut-through
- Fairness under N-to-1 incast
- Recovery from lost credit updates

Operational Gaps

- Limited tested-pair interoperability data
- Limited visibility into state and CtlOS frames

Standardized tests bridge protocol specifications and deployment.

LLR & CBFC Validation with SONiC Ranking Test

01 CONFIGURE

SONiC Ranking Test

- SAI set/get
- Capability checks
- Case orchestration & parameter sweeps

02 INJECT

Keysight Tester

- LLR / CBFC traffic
- Multi-flow / multi-VC / incast
- Fault injection & timestamping

03 OBSERVE

Device Telemetry

- SAI counters
- NACK / replay / credits
- State, where supported

04 VALIDATE

Test Reports

- Pass/fail vs. test criteria
- Tested-pair interoperability
- Configuration baselines

Cross-check tester measurements with device telemetry.

Test specification: SONiC Ethernet Scale-Up Test Standard | PR #2389

Test Framework and Test Insights

Confronting compliance and performance pain points, distilling rules for at-scale deployment, and driving open industry collaboration



LLR & CBFC Test Coverage & Core Validation Matrix

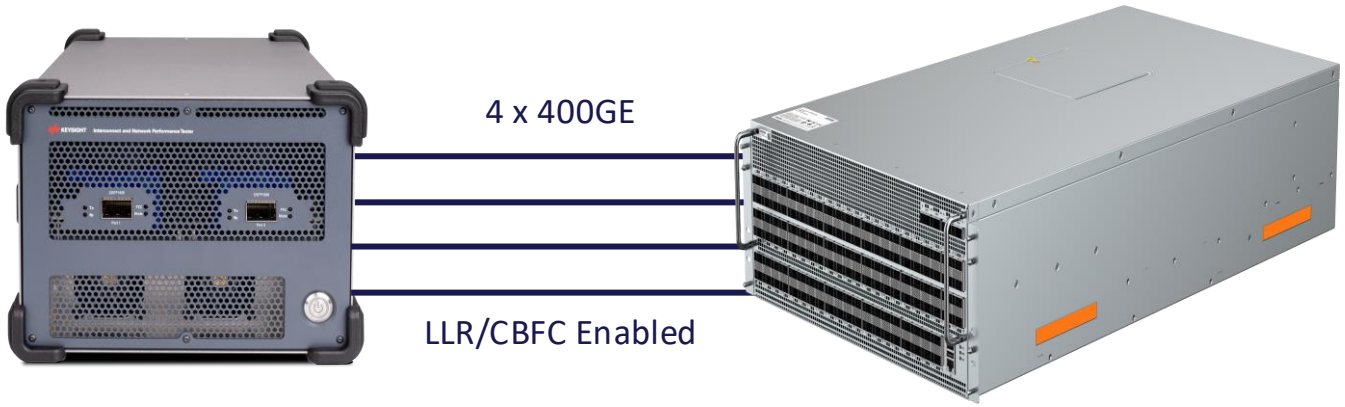
Covering user-facing functionality and field-deployment scenarios.

LLR Validation	CBFC Validation
✅ State Machine	✅ Credit Accounting
✅ Exception Recovery	✅ Backpressure Handling
✅ Poison / Flush Handling	✅ Traffic Isolation
✅ Buffer & Frame Boundaries	✅ Incast Convergence
✅ Mixed Traffic Coexistence	✅ Capacity & RTT Limits

End-to-End Lossless Ethernet Validation

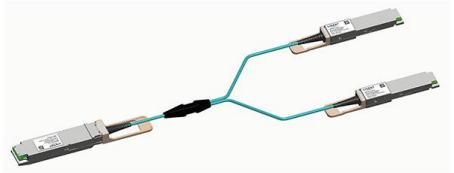
- Validated line-rate operation, zero-loss recovery , latency/Jitter, incast fairness, etc.
- Exercise min & max frame sizes, min & max link distance for sustained line-rate operation
- Exercise various function paths and error condition handling

LLR & CBFC Switch Test Topology



Keysight INPT 800GE

Alibaba Ethernet scale-up Switch



800G to 2x400G AOC 10m



LLR & CBFC Test Results

All 22 test cases passed

Case ID	Core Content	Result
LLR-01	Basic LLR Sequence/ACK	Pass
LLR-02	Retry (NACK)	Pass
LLR-03	Retry (Replay Timer Expired)	Pass
LLR-04	NAK Generation	Pass
LLR-05	Poison CRC Handling	Pass
LLR-06	OUTSTANDING_Seq_Max	Pass
LLR-07	Buffer Capacity	Pass
LLR-08	Flush Recovery	Pass
LLR-09	Packet Size Traversal	Pass
LLR-10	Mixed LLR and non-LLR Frames	Pass

Case ID	Core Content	Result
CBFC-01	Sender Credit Handling	Pass
CBFC-02	Receiver Credit Handling	Pass
CBFC-03	Credit Loss/Recovery	Pass
CBFC-04	Backpressure	Pass
CBFC-05	VC Mapping – VLAN PCP+DEI	Pass
CBFC-06	VC Mapping – IPv4 DSCP	Pass
CBFC-07	VC Mapping – IPv6 DSCP	Pass
CBFC-08	Per-VC Credit	Pass
CBFC-09	Per-Port Credit	Pass
CBFC-10	N-to-1 Incast: Lossless	Pass
CBFC-11	N-to-1 Incast: Lossless + Best Effort	Pass
CBFC-12	Buffer Capacity	Pass

LLR Sample Case Analysis

Frame Size (bytes)	Tx Frames	Rx Frames	Frames Delta	L1 Line Rate Usage (%)	Cut-Through Avg Latency (ns)	Cut-Through Min Latency (ns)	Cut-Through Max Latency (ns)
64	714286000000	714286000000	0	99.6161	301	300	316
65	705882000000	705882000000	0	99.6157	303	302	318
127	408163000000	408163000000	0	99.6162	303	301	318
128	405405000000	405405000000	0	99.6175	303	301	317
129	402685000000	402685000000	0	99.6179	302	300	317
255	218182000000	218182000000	0	99.6162	304	302	318
256	217391000000	217391000000	0	99.6171	305	304	318
257	216606000000	216606000000	0	99.6170	305	304	319
511	112994000000	112994000000	0	99.6142	300	299	315
512	112782000000	112782000000	0	99.6142	300	299	314
513	112570000000	112570000000	0	99.6139	301	301	317
1517	39037085230	39037085230	0	99.6118	307	305	320
1518	39011703511	39011703511	0	99.6115	305	304	318
1519	38986354775	38986354775	0	99.6123	305	304	320
2047	29027576197	29027576197	0	99.6145	305	303	317
2048	29013539651	29013539651	0	99.6147	305	303	318
2049	28999516674	28999516674	0	99.6148	305	303	320
4095	14580801944	14580801944	0	99.6137	302	302	316
4096	14577259475	14577259475	0	99.6122	303	302	316
4097	14573718727	14573718727	0	99.6118	305	304	317
9215	6497022198	6497022198	0	99.6120	319	318	330
9216	6496318752	6496318752	0	99.6118	317	316	329

No Link error

Tx Port	Rx Port	Traffic Item	Tx Frames	Rx Frames	Frames Delta	Loss %
Ethernet - 001	Ethernet - 002	LLR	1,853,055,916	1,853,055,916	0	0
Ethernet - 001	Ethernet - 002	Non LLR	97,529,258	97,529,258	0	0

Inject FEC error

Tx Port	Rx Port	Traffic Item	Tx Frames	Rx Frames	Frames Delta	Loss %
Ethernet - 001	Ethernet - 002	LLR	1,853,055,916	1,853,055,916	0	0
Ethernet - 001	Ethernet - 002	Non LLR	97,529,258	97,529,255	3	0

LLR-09: Frame size sweep: LLR traffic, no link error, 20M cable

- Line-rate throughput & zero loss across all frame sizes
- Switch forwarding latency 299~ 330ns, end-to-end latency (include cables & transceivers) 650~710 ns
- Jitter < 31 ns

LLR-10: Mixed LLR/non-LLR traffic, inject FEC error

- LLR traffic: no loss after replay, non-LLR traffic: some loss
- Comparable latency for LLR traffic and non-LLR traffic
- Latency to application increased over 1200 ns

CBFC Sample Case Analysis

L1 Bits Received	50380473024	50395381344	50428502208	146,373,923,4
L1 Bits Received Rate	67309716	67309899	67309582	39952955835
L1 Line Rate Transmit (%)	49.9478	49.9513	0.0167	0.0000
L1 Line Rate Receive (%)	0.0168	0.0168	0.0168	99.8823
Transmit Arp Gratuitous	0	0	0	0
Transmit Arp Reverse	0	0	0	0

Traffic distribution during 2:1 Incast congestion

L1 Bits Received Rate	67310044	67309841	67310033	39952961440
L1 Line Rate Transmit (%)	33.3042	33.3052	33.3064	0.0000
L1 Line Rate Receive (%)	0.0168	0.0168	0.0168	99.8824
Transmit Arp Gratuitous	0	0	0	0
Transmit Arp Reverse	0	0	0	0

Traffic distribution during 3:1 Incast congestion

Distance (One Way)	Frame Size (Bytes)	RTT (ns)	Credit Limit for Line-Rate Tput	Rx Rate
500 m	64	5,000	7,825	99.99%
500 m	9,216	5,000	6,240	99.65%
10 m	64	100	1,990	99.99%
10 m	9,216	100	1,875	99.65%

CBFC-11: 3:1 Incast, 1 lossless VC + 1 best effort VC

- Uniform bandwidth distribution across ports
- Effective credit management: zero loss for lossless VC, loss for best-effort VC under congestion

CBFC-12: Buffer Capacity (Supporting required RTT)

- Small frames carry higher credit overhead
- Credit limit scales with link distance; size it for small-frame line rate

LLR & CBFC Test Insights

Specification Compliance & Tuning

- **Out-of-spec behaviors requires tuning**
  - Poison CRC frame forwarding
  - Incast fairness
- **CtlOS bandwidth overhead**
  - ~0.38% under typical spacing
- **Port-level credit sharing limitation**
  - No VC isolation, suitable only for rapid deployment with single type of traffic flow
- **Deployment scenario-specific tuning**
  - Credit Limit, Outstanding_Data_Max / Outstanding_Seq_Max
  - Provision per Link distance & frame size for max Tput

Test Methodology & Equipment

- Link-layer testing depends on **hardware-based test platforms**
- **Link level error injection is critical** for compliance and boundary-condition testing
  - Inject various error types to exercise different function code/path
- Need to **capture and analyze the preamble** and **CtlOS** in PCS lane

LLR & CBFC Endpoint IP Test



UniVista Pre-silicon Simulation Platform

UniVista Scale-Up Speed Adapter

Keysight INPT 800GE

- LLR: All 10 test cases passed
- CBFC: 9 passed, 3 incomplete:
  - CBFC-07: IP does not yet support IPv6
  - CBFC-10/11: Incast cases — pending joint three-party testing

Call to Action

Open Test Reports

Published shared test data and performance baselines.

https://github.com/sonic-net/SONiC/pull/2596

Alibaba · Keysight · Broadcom · Univista

Open Test Framework | 2026 Q4

Open-source SONiC Ranking Test cases
and code on GitHub for repeatable validation.

Build the Ecosystem

Adopt the specification, contribute test cases,
and validate multi-vendor interoperability.

Extend Test Coverage

Expand to ESUN, rack-level testing, L3 routing, and
AI telemetry.

Thank You!

Q & A

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